

Power Electronics — Chapter 1: DC-DC Converters

POWER_E_466

1 Buck Converter

The buck converter is a DC-DC converter that steps down voltage ($V_{out} < V_{in}$).

1.1 Motivating the topology

Start from a black-box model: given $V_{in} = 50\text{ V}$, we want an average output of $V_{out} = 20\text{ V}$ across a resistive load.

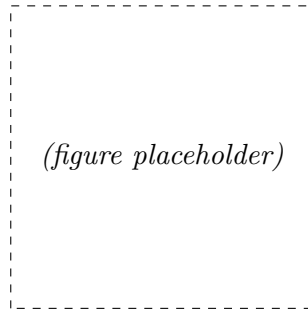


Figure 1: Black-box step-down converter: $V_{in} = 50\text{ V}$ in, unknown network, $\langle V \rangle = 20\text{ V}$ across the load.

The simplest way to reduce an average voltage is to switch the input on and off periodically (an ideal SPDT switch connecting the load either to V_{in} or to ground), so that the average value is controlled by the fraction of time spent connected to V_{in} — the *duty cycle* D :

$$\langle V \rangle = D V_{in}.$$

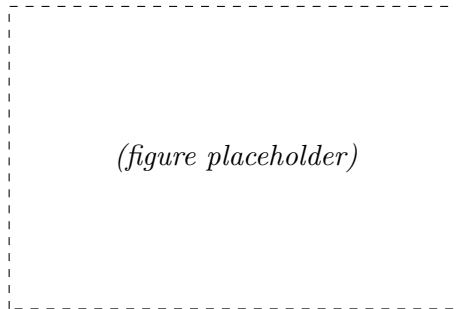


Figure 2: Ideal switch model: pole connects to V_{in} for a fraction D of the switching period T_s , and to ground for the remaining $(1 - D)T_s$, producing a PWM waveform whose average is $\langle V \rangle$.

1.2 Why we need an LC filter

The switched (PWM) waveform is not a clean DC voltage — it still contains switching-frequency ripple. From a Fourier-series point of view, the desired DC output corresponds only to the $f = 0$ Hz (DC offset) term; every other harmonic should be attenuated. A low-pass LC filter does exactly this: it passes the DC component while suppressing the switching-frequency content.

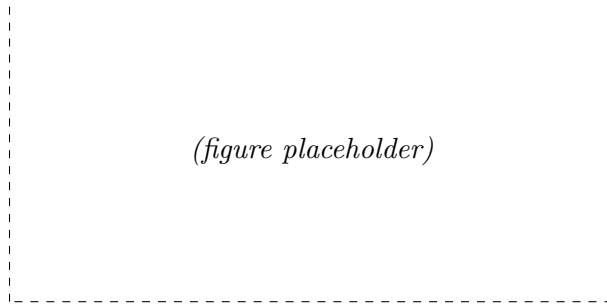


Figure 3: Buck converter: switch D followed by an LC low-pass filter, driving a resistive load. This is the buck converter topology.

1.3 Inductor and capacitor averaging identities

For an inductor,

$$v = L \frac{di}{dt} \iff i = \frac{1}{L} \int v dt,$$

so if i is periodic in steady state, the average inductor voltage must be zero:

$$\langle v_L \rangle = 0 \quad (\text{volt-second balance}).$$

For a capacitor,

$$i = C \frac{dv}{dt} \iff v = \frac{1}{C} \int i dt,$$

so if v is periodic in steady state, the average capacitor current must be zero:

$$\langle i_C \rangle = 0 \quad (\text{current-second/charge balance}).$$

1.4 Steady-state inductor current ripple

Over one switching period T_s , the inductor current i_L ramps up while the switch connects to V_{in} (duration DT_s) and ramps down while it connects to ground (duration $(1-D)T_s$):

$$\frac{di_L}{dt} = \frac{v_L}{L}, \quad 2\Delta i_L = \frac{V}{L} DT_s \quad (\text{rising segment}), \quad 2\Delta i_L = \frac{V}{L} (1-D)T_s \quad (\text{falling segment}).$$

1.5 Worked numerical example

Given $V_{in} = 50$ V, $D = 0.5$, $L = 10$ mH, $T_s = 1$ ms, and (for this simplified example) an output branch modelled as a current source with $V = 0$:

Switch ON ($t = 0^-$ to the switching instant).

$$\frac{di_L}{dt} = \frac{50}{10^{-3}} = 50 \times 10^3 \text{ A/s} \longrightarrow \Delta i_L = 5 \text{ A}.$$

At $t = 0^+$: $i_L = 5$ A, $i_O = 0$, $i_C = i_L - i_O = 5$ A.

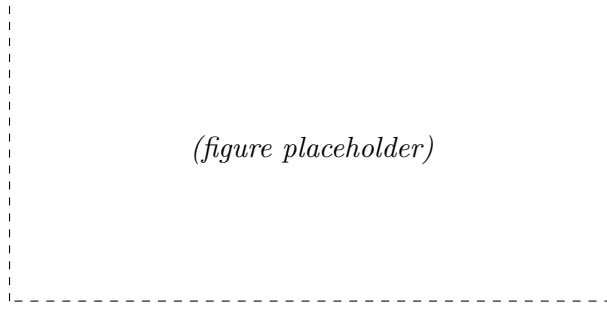


Figure 4: Steady-state inductor current: triangular ripple of peak-to-peak $2\Delta i_L$, rising for DT_s and falling for $(1 - D)T_s$ each switching period.

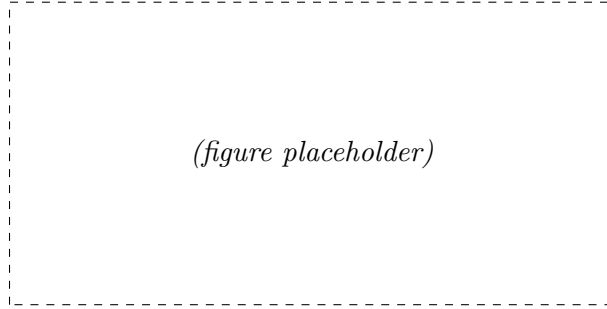


Figure 5: Buck circuit during the ON interval: $V_{in} = 50\text{ V}$, $D = 0.5$, $L = 10\text{ mH}$, $T_s = 1\text{ ms}$, output branch at $V = 0$.

Switch OFF.

$$\frac{di_L}{dt} = \frac{0}{10^{-3}} = 0\text{ A/s},$$

so i_L stays constant at 5 A while the switch is off (for this particular example, since the inductor sees 0 V in both states here).

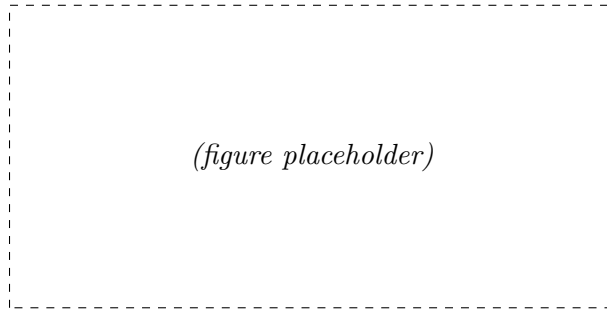


Figure 6: Buck circuit during the OFF interval, same component values as above.

Is this steady state? Not necessarily — V_{out} can still change from cycle to cycle if the circuit is not yet in continuous, periodic operation. As i_O increases from 0, $i_C = i_L - i_O$ decreases; by charge balance, $\langle i_C \rangle = 0$ is exactly the condition that defines the steady-state operating point.

1.6 Realizing the switch with real devices

The ideal SPDT switch (poles A , B , C) is not a physical component. It is realized using one *active* switch (a transistor, controlled by a gate/base signal) for the A – B throw, and one *passive*

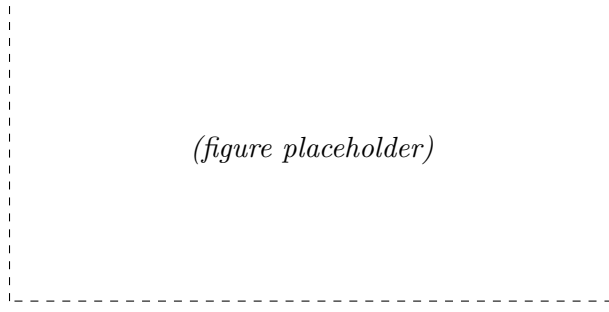


Figure 7: Buck circuit with increased output current ($V \approx 2.5\text{ V}$ example): as $i_O \uparrow$, $i_C \downarrow$, illustrating the approach to $\langle i_C \rangle = 0$ (charge balance).

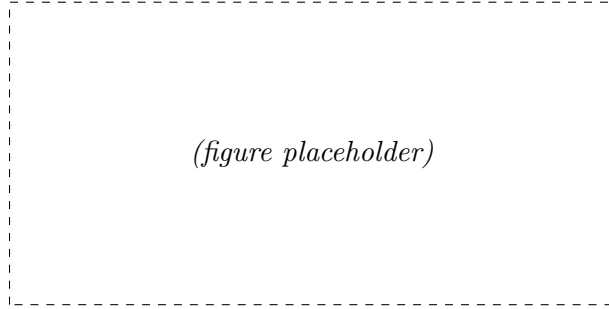


Figure 8: Classic multi-period inductor current waveform $i_L(t)$ ramping toward steady state: slope $\frac{V_g - v(t)}{L}$ while on, slope $\frac{-v(t)}{L}$ while off, converging so that $i_C \approx 0$ in steady state (cf. Erickson, *Fundamentals of Power Electronics*).

switch (a diode, which conducts automatically depending on the circuit's own voltage/current) for the C – B throw.

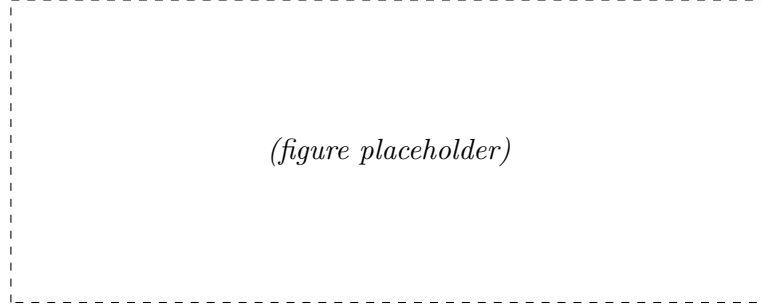


Figure 9: Replacing the ideal switch with a MOSFET (A – B) and a diode (C – B), giving the practical buck converter realization.

Node A–B (transistor position). When the ideal switch is ON: $V_{AB} = 0$, $I_{AB} = I_L$. When OFF: $V_{AB} = V_S$, $I_{AB} = 0$. This (I, V) locus (first quadrant only: on along the current axis, off along the voltage axis) matches an *active* switch such as a MOSFET/BJT/IGBT.

Node C–B (diode position). When the ideal switch is ON (meaning the A – B throw is closed): $V_{CB} = -V_S$, $I_{CB} = 0$. When OFF (meaning C – B conducts instead): $V_{CB} = 0$, $I_{CB} = I_L$. This locus (blocks negative voltage, conducts positive current, no external control terminal) matches a *passive* switch: a diode.

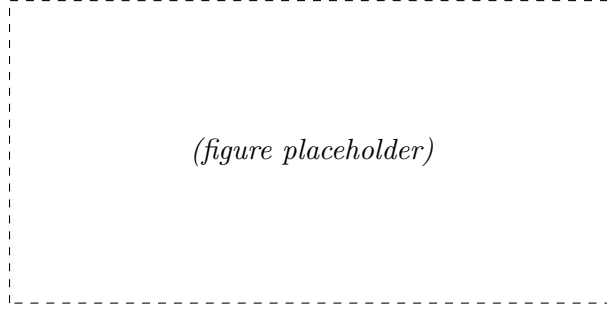


Figure 10: I_{AB} – V_{AB} switch locus for the transistor position: ON along the current axis, OFF along the voltage axis.

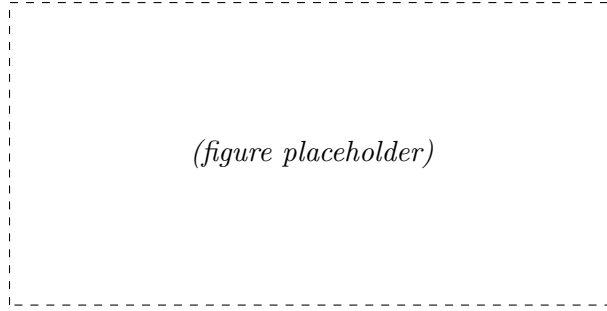


Figure 11: I_{CB} – V_{CB} switch locus for the diode position: blocks reverse voltage, conducts forward current.

1.7 Switch device characteristics

- **BJT / IGBT** are active switches: the conducting state is set entirely by the control-terminal signal, independent of the waveforms at the power terminals. Ideal characteristic: $i = 0$ when off (blocks $v \geq 0$); $v \approx 0$ when on (conducts $i \geq 0$). Poor/no reverse-conducting or reverse-blocking capability.
- **MOSFET** behaves similarly, but can additionally conduct in reverse. It is normally operated with $i \geq 0$, same as BJT/IGBT, so an active SPST switch can be realized with a BJT, IGBT, or MOSFET.
- **Diode** is a passive switch with no control terminal: its state is determined entirely by the terminal waveforms $i(t)$, $v(t)$. Ideal characteristic: off ($i = 0$) when $v < 0$; on ($v = 0$) when $i > 0$. It can block negative voltage but not positive voltage.

1.8 Summary

$$\boxed{V_O = D V_S} \quad \text{with } \langle i_C \rangle \rightarrow 0 \text{ in steady state.}$$

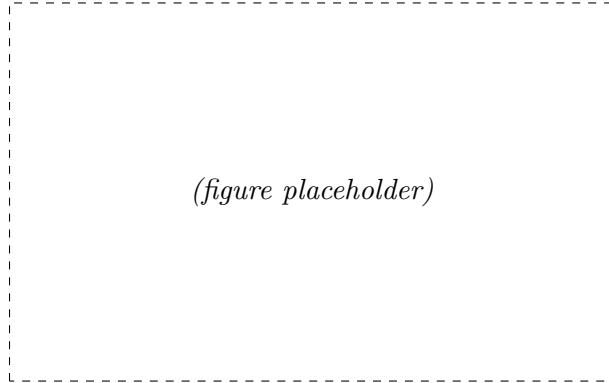


Figure 12: Ideal vs. real-world switch characteristics for BJT/IGBT, MOSFET, and diode (textbook reference figures).

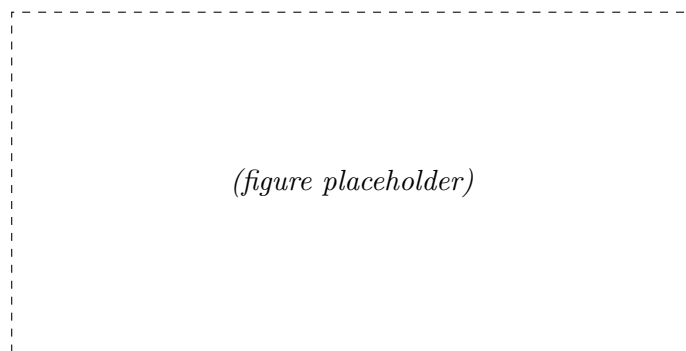


Figure 13: Final buck converter circuit: MOSFET (A–B), freewheeling diode (C–B), LC output filter, load.